AARTI KRISHAN KHATRI

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PROFESSIONAL SUMMARY

Computer Science student at Texas Tech University with hands-on experience in machine learning, data engineering, and scientific computing. Proven track record of building ML pipelines for spectral classification, automating laboratory processes, and developing data analysis tools for environmental and biomedical research. Seeking full-time roles in software engineering, data science, or AI/ML to apply analytical and programming skills to real-world challenges.

EDUCATION

Bachelor of Science in Computer Science, Honors Expected Dec 2026

Texas Tech University, Lubbock, TX | GPA: 3.8/4.0

Relevant Coursework: Data Structures & Algorithms, Machine Learning & AI, Data Science, Software Engineering, Databases, Distributed Systems, Operating Systems, Computer Networks

PROFESSIONAL EXPERIENCE

Research Assistant | *Dept. of Plant & Soil Sciences, Texas Tech University* Oct 2025 – Present

- Developed automated data analysis pipelines in Python (Pandas, NumPy, SciPy) to process LI-7815 gas exchange analyzer data, reducing manual analysis time by 80%.
- Engineered data parsing and segmentation tools to extract, clean, and structure raw instrument files across multiple soil treatment groups and experimental replications.
- Built linear regression models and statistical testing workflows to quantify CO_2 and H_2O flux rates, generating R^2 metrics and p-values for each treatment condition.
- Automated generation of formatted Excel reports and publication-quality visualizations (Matplotlib, OpenPyXL), delivering reproducible outputs for multi-treatment experimental data.

Machine Learning Research Assistant | *Srivastava Lab, Texas Tech University* Oct 2024 – Oct 2025

- Analyzed biomolecular spectral data of cancer exosomes and biomolecules, improving nanosensor sensitivity by 15% through signal processing techniques.
- Developed and optimized ML models (LDA, PCA-LDA, PCA-KNN) for spectral classification, achieving >70% prediction accuracy for disease diagnostics.
- Applied statistical techniques to enhance signal-to-noise ratios, improving data quality for SERS spectral analysis.
- Co-authored research accepted for presentation at BMES 2025: Biomineralized SERS Nanotags for Disease Diagnostics.

Data Science Intern | *CASFER, Case Western Reserve University* Jun 2025 – Jul 2025

- Mapped global soil nutrient distributions using Python and R (kriging interpolation, Geopandas, terra) with the WoSIS global database for geospatial visualization.
- Built and benchmarked ML models (Random Forest, XGBoost, Gradient Boosting, Ridge Regression) to predict peptide UV-Vis absorbance, streamlining experimental design.
- Applied optimized Random Forest model to 20,000+ peptide sequences, identifying top candidates for laboratory validation.

Software & Automation Intern | *CASFER, Texas Tech University* Jun 2024 – Jul 2024

- Automated 50+ laboratory processes using the Opentrons OT-2 robot, reducing manual workload by 90%.
- Developed Python-based liquid handling protocols with ±0.2 µL accuracy, increasing experimental reproducibility.
- Enhanced experimental data analysis workflows, improving nitrogen capture efficiency by 10%.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, SQL, R, C#, JavaScript

Libraries & Frameworks: NumPy, Pandas, Scikit-learn, SciPy, Matplotlib, XGBoost, Biopython, Geopandas, OpenPyXL, terra, sf

Tools & Platforms: Git, VS Code, JupyterLab, Google Colab, R Studio, Power BI, Microsoft Excel, Opentrons OT-2

Focus Areas: Machine Learning, Data Science, Data Engineering, Scientific Computing, Automation, Geospatial Analysis, Bioinformatics

AWARDS & HONORS

Presidential Merit Scholarship (2023–Present) | Gopal Lakhani & Zelda Doris Bond Scholarships (2025) | President's List (Spring 2023, 2024) | Dean's List (Fall 2024, 2025)